Pulmo-TS2ONN: A Novel Triple Scale Self Operational Neural Network for Pulmonary Disorder Detection Using Respiratory Sounds

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***Abstract*— Pulmonary disorders (PDs) are one of the substantial hazards to human life, which can be diagnosed by a variety of clinical modalities, including peak flowmeter and spirometry measurements, chest auscultation-based respiratory sound (RS) measurements, etc. Analyzing the acoustic RS measurements is one of the inexpensive yet essential diagnostic methods for identifying PDs as these RSs are correlated with structural flaws of the lungs that occur due to PDs. Additionally, the development of the digital stethoscope facilitates the continuous measurement of acoustic RSs of any individual, which can be exploited to identify a variety of PDs. In this article, we have proposed a triple time-frequency feature set driven triple-scale self-operational neural network (TS2ONN) architecture, namely Pulmo-TS2ONN, to classify a wide spectrum of PDs using the RSs. The proposed pulmo-TS2ONN comprises three major stages: preprocessing, triplet time-frequency feature set (TTFFS) extraction, and finally classification of seven class PDs by using TS2ONN architecture which utilizes the improved nonlinear neural backbone of self-operational neural network (SONN) in place of the linear neural architecture used in conventional deep learning (DL) networks. Upon experimental evaluation, the proposed framework outperforms the existing noteworthy research works by achieving the highest performance rates of 98.88%, 98.27%, and 99.84% for accuracy, sensitivity, and specificity, respectively. Lastly, the proposed framework is implemented on a quad-core ARM-A7-based Raspberry Pi-4 microcontroller, allowing the possibility of translating the research into real clinical situations for RS-based PD screening.**

***Index Terms*— Auscultation measurements, classification, pulmonary disorders (PDs), respiratory sounds (RSs), self-operational neural network (SONN).**

I. Introduction

PULMONARY disorders (PDs) have become one of the leading causes of global mortality, killing more than 8 million people across the globe [1]. As a result of air pollution and an aging population, the prevalence of PDs among the elderly and the general population has been increasing significantly each year [2]. One of the latest reports published by the world health organization has classified the following diseases as "Big Five" PDs, which are lower respiratory tract infection (LRTI), chronic obstructive pulmonary disease (COPD), asthma, tuberculosis (TB), and lung cancer [3]. Pulmonologists use different diagnostic modalities, such as spirometric [4] and peak flowmeter-based measurements [5], photoplethysmogram [6], FeNO tests [7], chest X-rays, etc., to assess the condition of the lungs. However, most of these modalities are either highly dependent on patient effort, laborious, or very costly [7]. However, respiratory sounds (RSs) heard during chest auscultation are still used as a key diagnostic modality to identify any abnormalities present in the lungs, as these RS characteristics change drastically if any anatomical flaw occurs in the lungs as a result of the manifestation of PDs [4], [8]. Wheeze and crackle are two of the most common adventitious lung sounds. Wheeze is a high-pitch adventitious event that can be observed in the expiration cycle of acoustic RS measurements if the air pathways get clogged by sputum, mucus, and other secretary substances [5] in case of COPD, and asthma. Whereas, crackle is an explosive and discontinuous sound that occurs in the inspiratory phases of the breathing cycle and is related to interstitial lung illnesses such as chronic bronchitis, pneumonia, and lung fibrosis [9]. Therefore, these RS measurements can be exploited for the identification of different respiratory anomalies of lungs [10]. In this regard, the digital stethoscopes enable continuous recording of the individuals' RS measures, which may then be used to identify a variety of PDs [10], [11], [12]. Therefore, creating automated algorithms based on artificial intelligence (AI) in conjunction with the RSs would be very helpful in the early diagnosis of a number of respiratory disorders.

A. Related Works on RS-Based PD Detection

Since the outburst of COVID-19, several scientific investigations into developing various cough and RS measurement-based audio-bio-markers to evaluate respiratory health have been conducted by researchers worldwide, as these sounds are highly connected with structural defects in the lungs [13]. However, the majority of these studies concentrate on

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diagnosing adventitious RSs such as ronchi, crackle, wheeze, etc., rather than PDs [11], [14], [15]. In recent years, only a few studies have employed RS to identify a wide spectrum of PDs [16], [17], [18] with the help of different machine learning (ML) and deep learning (DL) architectures.

Additionally, most of these works have used only the ICBHI challenge database [9] to evaluate their frameworks. The ICBHI database has a total of seven different disease categories (COPD, asthma, pneumonia, upper respiratory tract infection (URTI), bronchiectasis (BRON), bronchiolitis, and LRTI) along with a healthy category, and thereby these works can be classified into two categories: 1) three-class PD classification (healthy, chronic, nonchronic disease); and 2) six-class PD classification (asthma and LRTI excluded).

1) *Three-Class PD Classification:* Perna [18] used mel-frequency cepstral coefficient (MFCC) representation in conjunction with a standard convolutional neural network (CNN) architecture to classify the RSs into three categories: healthy, chronic (BRON, COPD, and asthma), and nonchronic disease classes (pneumonia, bronchiolitis, and URTI). By employing this technique, the authors achieved the highest classification rate of 82%. Pham et al. [11] exploited a CNN architecture with a mixture of multiple experts (CNN-MoEs), which was trained using a teacher-student learning framework, to categorize the RSs into three classes using the mel spectrogram time-frequency representation (TFR). Using this framework they achieved a specificity of 82% and a sensitivity of 96%. However, this three-class PD classification becomes less relevant when real-time AI-based diagnostic systems are considered, as the patient or the physician is more concerned with finding out the precise diagnosis, i.e., the particular PD that the individual is afflicted with [8] and [16].
2) *Six-Class PD Classification:* To make the RS-based PD classification problem more meaningful from the clinical perspective, Basu and Rana [19] proposed a gated recurrent unit (GRU) based recurrent neural backbone to categorize the RSs of the ICBHI database into six diseased groups, and achieved an accuracy of 95.67%. ICBHI database is an imbalanced database, having a large number of RS recordings in COPD class. To mitigate this class imbalance issue, García-Ordás et al. [20] investigated the possibility of using a variational autoencoder to strengthen the minority classes. Following this, a CNN model was trained to classify the RSs into six categories, which attained a specificity of 98.6% and a sensitivity of 98.8%. Similarly, Shuvo et al. [16] used the empirical mode decomposition (EMD) technique to decompose the RS into intrinsic mode functions (IMFs) and chose the IMF which has the highest correlation with the original RS. Thereafter, scalogram TFR was extracted from the chosen IMF and fed to standard lightweight CNN (LCNN) architecture which has four convolution layers, six dense layers, and 3.8 million trainable parameters. Finally, utilizing this architecture, the authors have achieved an accuracy of 98.7%, a specificity of 100.0%, and a sensitivity of 98.6%. Since asthma is one of the minority PD classes in the ICBHI database, none of the research took this into account for the PD classification task.

Tripathy et al. [21] used only the KAUH database to classify healthy, pneumonia, COPD, and asthmatic RSs using different ML classifiers such as support vector machine (SVM), multilayer perceptron (MLP), light gradient boosting machine (LGBM) in conjugation with the spectro-temporal features extracted from the empirical wavelet transform (EWT)-decomposed IMFs. However, they achieved an inferior accuracy rate of 74.15% for the four-class disease classification task. Apart from this, recently, a few researchers have combined the RSs from other publicly available databases (KAUH database [10], and RespiratoryDatabase@TR [12]) with the RSs of ICBHI database to create a custom larger database for PD classification. Shi et al. [17] combined the RSs of RespiratoryDatabase@TR and ICBHI to classify the asthmatic, healthy, pneumonia, and COPD cases by employing a two-stream transformer, multiresolution channel and spatial attention-based DL architecture. By employing this framework, the authors achieved 91.56% accuracy on the combined RS database.

Thereby, it is evident that the majority of studies either concentrate on a single database for PD classification or take into account a subset of the several PD classes to perform the classification task. Therefore, it is more relevant to exploit all publicly available databases to create a robust PD classification framework that covers a larger number of PDs. In this context, Roy and Satija [8] utilized all three public RS databases to perform a seven-class PD classification task (LRTI excluded) for the first time, by modifying the existing inception module of GoogLeNet architecture to a lightweight inception model, called RDLINet. By training the RDLINet architecture in conjunction with the mel spectrogram TFRs in an end-to-end fashion, the authors achieved the highest accuracy of 96.6%, a sensitivity of 96.2%, and a specificity of 98%, respectively.

B. Objective and Key Contribution

Due to the similarity of the symptoms across different PDs, it is challenging for a pulmonologist to diagnose the condition just based on the auscultation sounds [22]. Even a skilled doctor might make a mistake while making inferences from the auscultation due to the similarities between RSs [22] if they do not employ additional astute diagnostic methods. In general, it is advised to go for additional tests such as spirometric measures [12], peak-flowmeter-based measures [5], etc., to confirm the disease condition, however, several disadvantages are associated with these modalities as discussed earlier.

Therefore, the main objective of the proposed framework is to create an efficient DL framework in conjunction with different TFRs extracted from the RSs, which can be used as an assistive diagnostic modality for PD identification. In this article, we propose a triplet time-frequency feature set (TTFFS)-based triple scale self-operational neural network (TS2ONN), namely the Pulmo-TS2ONN for the efficient categorizing of PDs using the RSs.

Additionally, this work is a technical extension of our previous work published in 18th edition of IEEE International

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Symposium on Medical Measurements and Applications (MeMeA) [23], where we exploited the potential of the recently developed heterogeneous neural model of self-operational neural network (SONN) [24] for RS based asthma classification and showed that the SONN surpasses the performance of an existing homogeneous linear neural model of vanilla CNNs [23]. In this article, we have proposed a multiscale representation of the SONN to process and extract compact class-specific embedding representation from the TTFFS extracted from input RSs for seven-class PD classification. Further, we have implemented the proposed framework in a Raspberry Pi-4 microcontroller that enables the prospect of translating this research into a real-life clinical scenario for PD screening using RS measurements. To the best of our knowledge, this is the first work that utilizes the multiscale representation of the aforementioned nonlinear deep neural network architecture: SONN, for RS-based seven-class PD categorization. The major contributions of this research are.

1) Investigating the potential of triplet TFR-based TTFFS that contains mel spectrogram, constant-Q transform (CQT), and MFCC representations extracted from the RSs for the detection of PDs.
2) On the contrary to the existing DL models used in [8] and [23], etc., which exploit single stream neural design, we have developed a novel triple-scale TTFFS processing model by employing the multiscale representation of SONN architecture for the first time and fused their compact embedding repressions to classify the PDs efficiently.
3) Extensive evaluation of the proposed framework on the three publicly available databases and comparative analysis with existing vanilla CNN-based transfer learning networks in terms of different performance parameters.
4) On-device implementation of the proposed PD classification framework using a low-resource Raspberry pi-4 microcontroller.

The rest of the article is organized as: Section II describes a brief overview of the databases used to evaluate the classification performance of the proposed Pulmo-TS2ONN. Section III includes a detailed discussion on each of the stages of the proposed framework; Section IV discusses the performance evaluation and comparison of the proposed framework with the existing state-of-the-art (SOTA) techniques; and finally, Section V concludes the research.

II. Details of Database

This section describes the three publicly available RS databases: 1) ICBHI challenge database [9]; 2) King Abdullah University Hospital (KAUH) RS sound database [10]; and 3) RespiratoryDatabase@TR (RD@TR) [12], that have been utilized to develop the proposed Pulmo-TS2ONN framework for the detection of PDs. Table I provides a comprehensive description of all three databases, which includes demographic information of the subjects, acquisition devices used to collect the RSs, and various PD classes. In this research, we have combined the following PD classes such as asthma, COPD, BRON, pneumonia, bronchiolitis, URTI, and healthy, to create a bigger data repository for developing a seven-class PD categorization framework. Additionally, we have presented a brief description of the clinical background of each of the seven PDs in Table II.

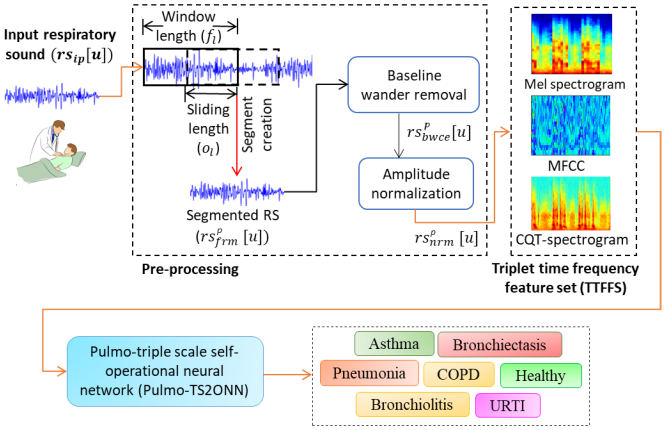


Fig. 1. Block diagram of the proposed Pulmo-TS2ONN framework for PD classification using RSs.

III. Proposed Framework

The proposed framework includes three major stages: 1) preprocessing of the input RS; 2) TTFFS creation; and 3) PD identification using the proposed triple-scale SONN (TS2ONN) architecture, namely the Pulmo-TS2ONN. Fig. 1 illustrates the block diagram of the proposed Pulmo-TS2ONN framework. Sections III-A–III-C include a thorough explanation of each of the aforementioned major stages.

A. Preprocessing

The preprocessing stage begins with resampling the input RS ($\text{rs}_{\text{ip}}[u]$) to 4 kHz rate [8], [14] and then framing the resampled RS ($\text{rs}_{\text{re}}[u]$) into 5 s length frames (f_l) while maintaining an overlap (o_l) of 50% with the subsequent frame

$$\text{rs}^p_{\text{frm}}[u] = \text{rs}_{\text{re}}\big[f_l \cdot p \cdot o_l + u\big], \quad p = 0, 1, 2, \ldots, P \tag{1}$$

where $\text{rs}^p_{\text{frm}}[u]$ and P denote the pth segmented frame of resampled RS, and the total number of segmented frames respectively. Thereafter, a discrete Fourier transform (DFT) based low-frequency baseline wandering component (BWC) 0–1 Hz eliminating filter [29] is employed on each of the segmented frames ($\text{rs}^p_{\text{frm}}[u]$). The DFT index ($k$) for f Hz frequency is calculated as $k = \lceil (f \cdot U/f_s) \rceil$, where f_s and U refer to the sampling frequency of lung sound signal, and the total number of DFT points in the segmented frame ($\text{rs}^p_{\text{frm}}[u]$), respectively. Thereafter, the DFT indices corresponding to BW noise are set to zero. Therefore, the BWC eliminated signal ($\text{rs}^p_{\text{bwce}}[u]$) can be obtained by taking the inverse DFT of the thresholded DFT vector ($\tilde{\text{RS}}^p_{\text{frm}}[k]$). The mathematical foundation for BWC elimination is described as [29]: The DFT of the pth segmented frame ($\text{rs}^p_{\text{frm}}[u]$) is calculated as

$$\text{RS}^p_{\text{frm}}[k] = \sum_{u=0}^{U-1} \text{rs}^p_{\text{frm}}[u] e^{\frac{-j2\pi \text{U.K.}}{U}} \tag{2}$$

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TABLE I
BRIEF DESCRIPTION OF THE THREE PUBLICLY AVAILABLE DATABASES

Database	Sampling frequency	Acquision device	Total number of subjects (male, female)	Age distribution in years (mean ± std. dev.)	Recording length	PD categories utilized in our study (number of RS recordings)
ICBHI [9]	4 kHz, 10 kHz, 44.1 kHz	Littmann 3200M, littmann classic II, welchallyn meditron, AKG C417L microphone	126 (79, 47)	43.0 ± 32	10 sec to 90 sec	Healthy (35), COPD (793), BRON (16), bronchiolitis (13), URTI (23), asthma (1), pneumonia (37), LRTI (2)
KAUH [10]	4 kHz	Littmann 3200M	112 (69, 43)	50.5 ± 19.4	15 sec to 50 sec	Healthy (105), fibrosis (5), BRON (18), pneumonia (15), COPD (33), asthma(98), heart failure (57), pleural effusion (5)
Respiratory Database@TR [12]	4 kHz	Littmann 3200M	41 (34,7)	38 to 68 years	at least 17 sec	COPD0 (60), COPD1 (60), COPD2(84), COPD3 (84), COPD4 (192)

TABLE II
BRIEF DESCRIPTION OF THE DIFFERENT PDS

PD category	Type of PD (chronic/ non-chronic)	Symptoms	Observable lung sounds (freq. range)	Diagnostic modality for disease identification
Asthma [25]	Chronic	Shortness of breath, chest pain, coughing, expiratory wheezing	Wheeze [26] (100-1000 Hz)	Spirometry, peak flow meter test
COPD [25]		Persistent coughing, wheezing, frequent lung infections	Wheeze	Spirometry or lung function test
BRON [25]		Coughing up yellow/ green mucus, fever, shortness of breath	Crackle [27] (100-200 Hz)	High-resolution CT (HRCT) scan
Pneumonia [25]	Non-chronic	Shallow breathing, stabbing chest pain, fever, loss of appetite	Crackle	Chest X-ray, CT scan, sputum test
Bronchitis [28]		Runny nose, fever, fatigue	Wheeze	Spiromtery, CT scan
URTI [25]		Sneezing, sore throat, headache, high fever, cough	Wheeze, stridor [26] ($>$ 500 Hz)	Nasal swab, sputum test

$$\mathrm{rs}^{p}_{\mathrm{bwce}}[u] = \frac{1}{U}\sum_{k=0}^{U-1} \tilde{\mathrm{RS}}^{p}_{\mathrm{frm}}[k] e^{\frac{j2\pi \mathrm{U.K.}}{U}} \tag{3}$$

where $\tilde{\mathrm{RS}}^{p}_{\mathrm{frm}}[k]$ denotes the thresholded DFT index vector derived from $\mathrm{RS}^{p}_{\mathrm{frm}}[k]$. $\tilde{\mathrm{RS}}^{p}_{\mathrm{frm}}[k] = [0, 0, \ldots, 0, \mathrm{RS}^{p}_{\mathrm{frm}}[k+1], \ldots, \mathrm{RS}^{p}_{\mathrm{frm}}[U-k-1], 0, \ldots, 0, 0]$. Lastly, the $\mathrm{rs}^{p}_{\mathrm{bwce}}[u]$ signal is normalized between the range of $[-1, 1]$ by employing the following equation: $\mathrm{rs}^{p}_{\mathrm{nrm}}[u] = (\mathrm{rs}^{p}_{\mathrm{bwce}}[u] / \max|\mathrm{rs}^{p}_{\mathrm{bwce}}[u]|)$, where $\mathrm{rs}^{p}_{\mathrm{nrm}}[u]$ denotes the pth normalized frame of RS. Additionally, we have adopted several data augmentation techniques, such as time stretching, noise addition, pitch shifting, etc., [8], to increase the number of samples in the minority class (all the PD classes except COPD [8]).

B. Triplet Time-Frequency Feature Set (TTFFS)

According to the literature, analyzing the RS signal is highly difficult owing to its high nonstationary character, which contains fluctuating frequency components over time [8]. Therefore, it is essential to transform the signal from one domain to another to obtain the comprehensive information present in the signal. In this aspect, transforming the time-domain RS to the time-frequency domain will help in capturing the time-varying spectral characteristics of the signal [11]. In this article, we have used an exhaustive set of TFRs from each of the normalized RS segments ($\mathrm{rs}^{p}_{\mathrm{nrm}}[u]$), such as: 1) mel spectrogram; 2) constant Q transform (CQT) spectrogram; and 3) MFCCs, to create the TTFFS.

1) Mel Spectrogram: Firstly, the short-time Fourier transform (STFT) [30] ($\mathcal{S}^{p}_{\mathrm{rs}}[h, f]$) is calculated from each of the normalized RS segments ($\mathrm{rs}^{p}_{\mathrm{nrm}}[u]$) using the following equation [23]:

$$\mathcal{S}^{p}_{\mathrm{rs}}[h, f] = \sum_{u=0}^{U-1} \mathrm{rs}^{p}_{\mathrm{nrm}}[u] \cdot \mathcal{W}[u - h\mathcal{H}] \cdot e^{-j\frac{2\pi u f}{U}} \tag{4}$$

$$\mathcal{W}[u] = 0.54 - 0.46 \cdot \cos\left(\frac{2\pi u}{\mathfrak{w}_l - 1}\right) \tag{5}$$

where $\mathcal{W}[u]$ is the Hamming window with 1024 samples ($\mathfrak{w}_l$), with hop-length ($\mathcal{H}$) of 524 samples. Thereafter, we project the Hertz frequency (f) to mel-scale frequency (f_{mel}) for constructing the mel-filter banks by employing the following equation [31]:

$$f_{\mathrm{mel}} = 2595 \cdot \log\left(1 + \frac{f}{700}\right). \tag{6}$$

In this work, we have considered 64 mel-bands. To extract the mel-spectrogram ($\mathcal{X}_{\mathrm{MelSpec}}$), the triangular mel-filters are multiplied with each STFT frame ($\mathcal{S}^{p}_{\mathrm{rs}}[h, f]$) [31].

2) CQT Spectrogram: CQT [32], [33] is one of the most commonly used audio TFRs [11] which is utilized for music analysis [32], adventitious RS event detection [11], etc. It imitates the way the human hearing system functions. CQT comprises Z number of filters which have a center frequency of f_z and a bandwidth of Ω_z, respectively. The mathematical relation between the bandwidth of the filters can be expressed as [32]: $\Omega_z = (2^{1/Z})^z \cdot \Omega_{\min}$, where $\Omega_{\min}$ and Z represent the filter bandwidth containing the lowest center frequency, and the number of filters per octave, respectively. Additionally, the quality factor ($\mathcal{Q}$) can be expressed as [32]: $\mathcal{Q} = f_z / \Omega_z$. The $\mathcal{Q}$-factors are constant across all frequency bins. Suppose the sampling frequency is f_s, then the window length ($\mathfrak{w}_l$) of each bin can be expressed as a function of the index of the bin (z)

$$\mathfrak{w}_l(z) = \frac{f_s}{\Omega_z} = \mathcal{Q}\frac{f_s}{f_z} \tag{7}$$

where (f_s/f_z) illustrates the total number of samples analyzed in each cycle at f_z. Therefore, the frequency in digital domain

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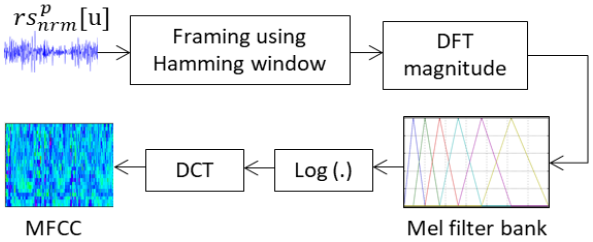


Fig. 2. MFCC extraction process from the normalized RS.

turn out as $(2\pi\mathcal{Q}/\mathfrak{w}_l(z))$. In this article, to extract the CQT, we have used the Hamming window. Thereby, the CQT can be computed as [32]

$$\mathrm{CQT}^p_{\mathrm{rs}}[z,m]=\frac{1}{\mathfrak{w}_l(z)}\sum_{u=0}^{\mathfrak{w}_l(z)-1}\mathrm{rs}^p_{\mathrm{nrm}}[u]\cdot\mathcal{W}[z,u-m]\cdot e^{-j\frac{2\pi\mathcal{Q}u\omega}{\mathfrak{w}_l(z)}}. \tag{8}$$

Finally, we take the squared magnitude of the $\mathrm{CQT}^p_{\mathrm{rs}}[z,m]$ to get the CQT spectrogram TFR ($\mathcal{X}_{\mathrm{CQT}}$).

3) MFCC: MFCC [34] is one of the most widely used audio feature representations for RS analysis [11]. Fig. 2 illustrates the block diagram of the MFCC extraction process. To extract the MFCC feature, first the normalized RS ($\mathrm{rs}^p_{\mathrm{nrm}}[u]$) is segmented into small frames ($\mathrm{rs}^i[u]$) with 1024 samples with 50% overlap, and a Hamming window ($\mathcal{W}[u]$) is applied on each of the frames as [34]: $y^i[u]=\mathrm{rs}^i[u]\cdot\mathcal{W}[u]$, where $y^i[u]$ denotes the transformed output signal generated from the corresponding frame $\mathrm{rs}^i[u]$. Thereafter, the DFT-based squared magnitude spectrum ($\tilde{Y}^i[k]$) is computed for each of the transformed output ($y^i[u]$). Mathematically, this process can be expressed as

$$\tilde{Y}^i[k]=\left|\sum_u y^i[u]e^{\frac{-j2\pi\mathrm{U.K.}}{U}}\right|^2. \tag{9}$$

These magnitude spectrum frames are then passed through the triangular mel-filter banks, which are defined by a constant mel-frequency (f_{mel}) interval [34]. Finally, a log transformation followed by a discrete cosine transform (DCT) is applied to the mel-filter bank projected magnitude spectrum to select the MFCC coefficients [34]. In this article, we have considered 30 mel-filters to construct the MFCC feature ($\mathcal{X}_{\mathrm{MFCC}}$).

The temporal visualization, mel spectrogram, CQT spectrogram, and MFCC TFR visualization of RSs from healthy subject is shown in Fig. 3(a)–(d), for BRON affected case in Fig. 3(e)–(h), for bronchiolitis in Fig. 3(i)–(l), for asthmatic case in Fig. 3(m)–(p), for pneumonia affected subject in Fig. 3(q)–(t), for URTI in Fig. 3(u)–(x), and lastly for COPD affected subject in Fig. 3(y)–(β).

C. Model Architecture of the Proposed Pulmo-TS2ONN

In this subsection, we discuss the architectural details of the proposed Pulmo-TS2ONN model utilized for PD classification. In contrast to our previously published work: AsTFSONN [23], where the neural network comprises only a single stream operational neural network design to extract features from

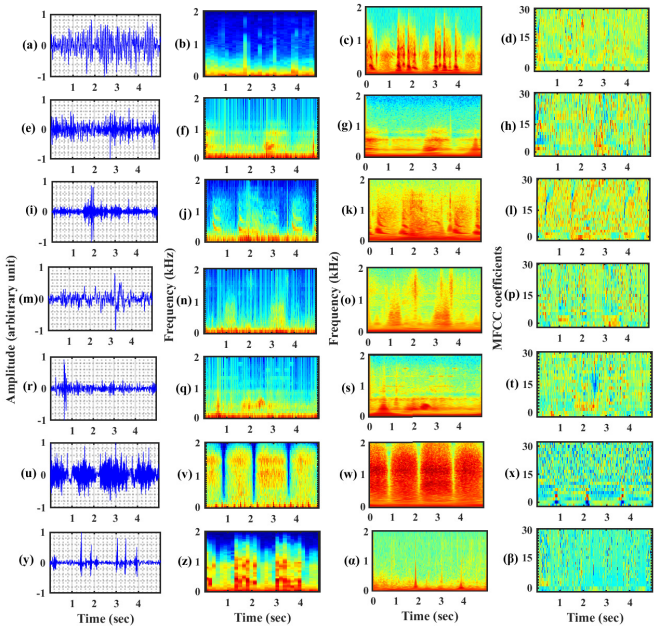


Fig. 3. Illustrates the temporal visualization, mel spectrogram, CQT spectrogram, and MFCC TFRs of (a)–(d) healthy subject's RS, (e)–(h) BRON, (i)–(l) Bronchiolitis, (m)–(p) Asthma, (q)–(t) Pneumonia, (u)–(x) URTI, and (y)–(β) COPD affected subject's RS.

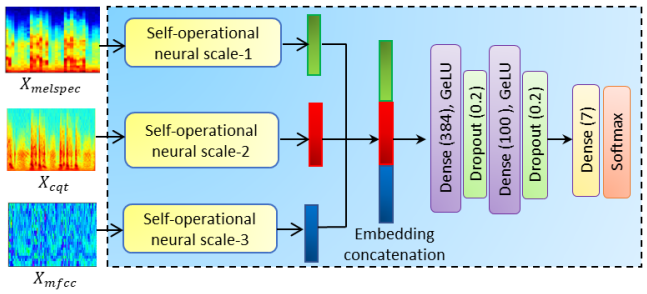


Fig. 4. Illustrates the internal architecture of the proposed Pulmo-TS2ONN model.

mel spectrogram TFR, the proposed Pulmo-TS2ONN model uses three self-operational neutral scales to extract compact feature representation from each of the three TFRs present in the TTFFS. In contrast to the standard CNN, which utilizes a linear perceptron model to extract features from the input tensor, the proposed self-operational neural scales (SONSs) employ the generative operational perceptron model (GOPM) of the SONN framework [24] to increase nonlinearity into the neural model [23]. Figs. 4 and 5 illustrate the model architecture of the proposed Pulmo-TS2ONN and SONS block, respectively. The details of each block used to construct the Pulmo-TS2ONN model are described below.

1) Self-Operational Neutral Scales: The main backbone of the SONS comprises the SONN layer, tan*h* activation layer, max-pooling (MP) layer, and 2-D global average pooling layer (GAP).

1) *SONN Layer:* The limitation of vanilla operational neural networks (VONNs) [35], is the requirement for a predefined library of operators and the need to search within that library to discover the ideal set of operators,

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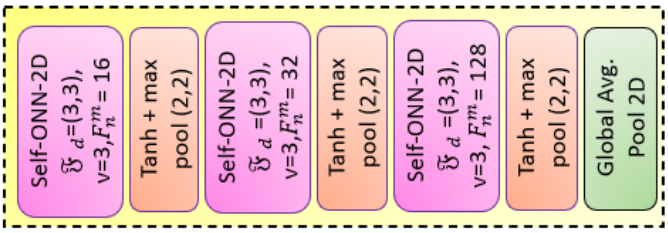


Fig. 5. Illustrates internal architecture of SONS.

are addressed by the GOPM designed for SONN [36]. Due to the excellent nonlinearity and reduced computational cost, SONNs outperform classic VONNs [24], [36]. Let us consider, $\mathcal{X} \in \Re^{\mathcal{A}\times\mathcal{B}\times\mathcal{C}_{\text{in}}}$ represents an input tensor with height $\mathcal{A}$, width $\mathcal{B}$, and depth $\mathcal{C}_{\text{in}}$. Additionally, $\mathfrak{F}_{\text{Cout}}$ indicates the $\mathcal{C}_{\text{out}}^{\text{th}}$ kernel of a layer. The traditional perceptron model of CNN is homogeneous and linear as it uses simple dot product-based multiplication and addition operations, namely the convolution operation [35]. In this case, the input tensor $(\mathcal{X})$ is convoluted with $\mathfrak{F}_{\text{Cout}}$ which is followed by the addition of a bias term $(\mathfrak{B}_{\text{Cout}})$. This process can be mathematically described as

$$\begin{aligned}\mathcal{X}_{\text{Cout}} &= \sum_{a,b,c_{\text{in}}=1}^{\mathcal{A},\mathcal{B},\mathcal{C}_{in}} \mathfrak{F}_{\text{Cout}}(a,b,c_{\text{in}})\cdot\mathcal{X}(a,b,c_{\text{in}}) + \mathfrak{B}_{\text{Cout}} \\ &= \mathfrak{F}_{\text{Cout}}^{T}\cdot\mathcal{X} + \mathfrak{B}_{\text{Cout}} \end{aligned} \tag{10}$$

where $\mathcal{X}_{\text{Cout}} \in \Re^{\mathcal{A}\times\mathcal{B}\times\mathcal{C}_{\text{out}}}$ denotes the output feature map which is then passed through some activation function to introduce the nonlinearity. To enhance the nonlinearity of the conventional perceptron model of the CNNs, Kiranyaz et al. [35] introduced VONN architecture, comprising a new perceptron model, namely the GOPM, which can exploit different sorts of nonlinear mathematical operators while processing any input features from a predefined library of such operators. Thus, the operation of VONN can similarly be written as [35]

$$\mathcal{X}_{\text{Cout}} = \beta\{\mathfrak{F}_{\text{Cout}}, \mathcal{X}\} + \mathfrak{B}_{\text{Cout}} \tag{11}$$

where β operator indicates the predefined dictionary of several nonlinear mathematical functions sinusoid (sin{·}), cosine (cos{·}), hyperbolic (sinh{·}, tanh{·}), exponential (exp{·}), etc. [35]. Since this β operator is not fixed, we must learn it during the training phase, which necessitates an extremely time-consuming search procedure. As a result, the training time increases to a great extent [35]. The VONN layer turns into a standard CNN layer when the β operator is a dot product.

However, in the case of SONN, a novel generative neural backbone is used which can create any arbitrary nonlinear function for each kernel element using a truncated Mac-Laurin series expansion during the training phase, which helps to mitigate the searching process of the nonlinear functions from a predefined dictionary and the high training time issue of VONN [23]

$$\begin{aligned}&\beta\left(\mathfrak{F}_{\text{Cout},1}, \mathfrak{F}_{\text{Cout},2}\ldots, \mathfrak{F}_{\text{Cout},v}, \mathcal{X}\right) \\ &= \sum_{v=1}^{V} \mathfrak{F}_{\text{Cout},v}^{T}\cdot\mathcal{X}^{v} \\ &= \mathfrak{F}_{\text{Cout},1}^{T}\mathcal{X} + \mathfrak{F}_{\text{Cout},2}^{T}\mathcal{X}^{2} + \ldots, + \mathfrak{F}_{\text{Cout},V}^{T}\mathcal{X}^{V} \end{aligned} \tag{12}$$

where $\mathcal{X}^v$ is vth power of the input tensor $\mathcal{X}$ and $\mathfrak{F}_{\text{Cout},v}$ denotes the learnable kernel weights corresponding to each $\mathcal{X}^v$, which will be learned by gradient descent based backpropagation scheme. Therefore, the GOPM present in the SONN transforms the input tensor by the following equation:

$$\mathcal{X}_{\text{Cout}} = \sum_{v=1}^{V} \mathfrak{F}_{\text{Cout},v}^{T}\cdot\mathcal{X}^{v} + \mathfrak{B}_{\text{Cout}}. \tag{13}$$

In this article, we have selected the SONN as the main building element of the SONS block due to its neural architecture's (GOPM) capability to produce improved nonlinear representations from the input tensors over traditional CNNs and VONNs.

2) *Tanh Activation Layer:* The goal of the activation function is to further nonlinearize the SONN output $(\mathcal{X}_{\text{Cout}})$. In this article, we have employed tan$h$ nonlinear activation [24], which is mathematically described as

$$\tanh(\mathcal{X}_{\text{Cout}}) = \frac{e^{\mathcal{X}_{\text{Cout}}} - e^{-\mathcal{X}_{\text{Cout}}}}{e^{\mathcal{X}_{\text{Cout}}} + e^{-\mathcal{X}_{\text{Cout}}}}. \tag{14}$$

3) *MP Layer:* The activation block's output is subsequently passed through the 2-D-MP block to reduce the dimensionality and capture the local features by retaining the maximum value of the feature map present in a window with a specified pool size, which is moved with the specified stride [8].

4) *GAP Layer:* The GAP layer is used to generate the final 1-D output embedding from each of the TFRs, processed through the three cascaded SONN layers present in SONS. GAP layer reduces the trainable parameter size and prevents over-fitting [37].

In this article, we have used three scales of SONS and each of the SONSs further contains a cascade of three SONN layers, followed by a hyperbolic tangent (tan$h$) nonlinear activation and a max-pooling layer after each of the SONN layers, as shown in Fig. 5. Finally, a 2-D GAP [37] is employed to extract a 1-D embedding output vector as the final outcome of SONSs. Therefore, the series of cascaded operations employed in ith SONS $(\zeta^{\text{SONS}_i}\{\cdot\})$ can be formalized from Fig. 5 as

$$\begin{aligned}&\zeta^{\text{SONS}_i}\left\{\mathcal{X}_{\text{TTFFS}_i}\right\} \\ &= \zeta^{\text{GAP}}\left\{\zeta^{\text{MP}}\zeta^{\tanh}\zeta^{\text{SONN}}\left\{\zeta^{\text{MP}}\zeta^{\tanh}\right.\right. \\ &\quad\left.\left.\times\zeta^{\text{SONN}}\left\{\zeta^{\text{MP}}\zeta^{\tanh}\zeta^{\text{SONN}}\left\{\mathcal{X}_{\text{TTFFS}_i}\right\}\right\}\right\}\right\} \end{aligned} \tag{15}$$

where $\mathcal{X}_{\text{TTFFS}_i}$, ζ^{SONN}, $\zeta^{\tanh}$, ζ^{MP}, and ζ^{GAP} denote the ith TFR element present in the TTFFS, and the functional operation of the SONN layer, tan$h$ activation layer, the MP layer, and lastly, the GAP layer, respectively. In this study, three SONN layers have a uniform filter dimension $(\mathfrak{F}_d)$ of 3 × 3, with filter number (F_n^m) of 16, 32, and 128, respectively. Additionally,

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TABLE III
DESCRIPTION OF THE TOTAL LEARNABLE PARAMETER SIZE OF THE PULMO-TS2ONN

Block name	Sub layers	Output dimension	Trainable parameters
SONS 1	Mel spectrogram input layer	$64 \times 39 \times 1$	0
	SONN + tanh + maxpool	$31 \times 18 \times 16$	448
	SONN + tanh + maxpool	$14 \times 8 \times 32$	13,856
	SONN + tanh + maxpool	$6 \times 3 \times 128$	110,720
	GAP 2D	128×1	0
SONS 2	CQT input layer	$64 \times 40 \times 1$	0
	SONN + tanh + maxpool	$31 \times 19 \times 16$	448
	SONN + tanh + maxpool	$14 \times 8 \times 32$	13,856
	SONN + tanh + maxpool	$6 \times 3 \times 128$	110,720
	GAP 2D	128×1	0
SONS 3	MFCC input layer	$30 \times 40 \times 1$	0
	SONN + tanh + maxpool	$14 \times 19 \times 16$	448
	SONN + tanh + maxpool	$6 \times 8 \times 32$	13,856
	SONN + tanh + maxpool	$2 \times 3 \times 128$	110,720
	GAP 2D	128×1	0
Embedding concatenation	Concat + GeLU + dropout	384×1	0
FCDL 1	Dense + GeLU + dropout	100×1	38500
Classification layer	Dense + Softmax	7×1	707
Total learnable parameters			**414,279**

we have utilized the Mac-Laurin series truncation factor (v) of 3 for the SONN layers. Finally, the GAP layer is used to generate the 1-D output embedding from each of the TFRs, processed through the three cascaded SONN layers present in each of the SONS. Thereby, the 1-D-embedding generation process from each of the TFR elements present in the TTFFS using SONSs can be mathematically described as

$$E^{\text{MelSpec}} = \zeta^{\text{SONS}_1}\{\mathcal{X}_{\text{MelSpec}}\} \tag{16}$$

$$E^{\text{CQT}} = \zeta^{\text{SONS}_2}\{\mathcal{X}_{\text{CQT}}\} \tag{17}$$

$$E^{\text{MFCC}} = \zeta^{\text{SONS}_3}\{\mathcal{X}_{\text{MFCC}}\} \tag{18}$$

where $\{E^{\text{MelSpec}}y, E^{\text{CQT}}, E^{\text{MFCC}}\} \in \Re^{128\times 1}$ denote the 1-D-embedding output generated from the triplet TFR set using SONSs.

Thereafter, these 1-D embeddings are concatenated with each other to form another fused 1-D vector ($E_{\text{concat}} \in \Re^{384\times 1}$) using the following method: $E_{\text{concat}} = [E^{\text{MelSpec}}; E^{\text{CQT}}; E^{\text{MFCC}}]$, which is then passed through the Gaussian error linear unit (GeLU) activation layer [38] and dropout layer with a dropout factor of 0.2. Following that, we utilize a fully connected dense layer (FCDL) with 100 neurons, GeLU activation, dropout rate of 0.2. The mathematical description for these dense layers is given below [38]

$$\mathcal{Z}_{\text{fc}} = \text{GeLU}(\mathcal{X}_{\text{fc}} \cdot \mathcal{W}_{\text{fc}} + \mathcal{B}_{\text{fc}}) \tag{19}$$

where $\mathcal{X}_{\text{fc}}, \mathcal{Z}_{\text{fc}}, \mathcal{W}_{\text{fc}}$, and $\mathcal{B}_{\text{fc}}$ denote the input-output vector, weights, and biases of the FCDL, respectively. Lastly, a classification layer with seven neurons and softmax activation [8] is used to get the class-specific probability values. Table III describes the total number of learnable parameters required to construct the Pulmo-TS2ONN model. Finally, the Pulmo-TS2ONN model is trained using a gradient descent-based weight updation approach by minimizing the categorical cross-entropy (CCE) loss function [8].

TABLE IV
OPTIMAL SIMULATION PARAMETERS OF PULMO-TS2ONN

Simulation parameter	Details
Learning rate	0.005
Optimizer	Adam
Epochs	250
Mel spectrogram dimension	64×39
CQT dimension	64×40
MFCC dimension	30×40
Batch size	128

IV. RESULT AND DISCUSSION

In this section, we compare the quantitative and qualitative performance of the proposed Pulmo-TS2ONN framework with some of the noteworthy prior works on RS-based seven-class PD classification.

A. Performance Parameters

To evaluate the performance of the proposed framework, we have used several standard performance metrics, such as accuracy (acu) [8], [39] specificity (spc) [39], sensitivity (sns) or recall (rcl) [8], precision (prc) [39], and $F1$-score [39]. All of these metrics can be calculated from the below-mentioned confusion matrix obtained while testing the framework [40]. Let us consider, the (7×7) confusion matrix for seven class PD classification is given as:

$$\begin{bmatrix} C_{1,1} & C_{1,2} & \cdots & C_{1,7} \\ C_{2,1} & C_{2,2} & \cdots & C_{2,7} \\ \vdots & \vdots & \ddots & \vdots \\ C_{7,1} & C_{7,2} & \cdots & C_{7,7} \end{bmatrix}$$

where $C_{m,m}$ corresponds to the correctly categorized sample, and $C_{m,n}(m \neq n)$ refers to the samples that belong to the mth PD class, however incorrectly classified as nth PD class. Therefore, the equations of the metrics for the mth PD class are provided as [40] and [39]: $\text{acu}(\%) = (\text{TP}^m + \text{TN}^m/\text{TP}^m + \text{TN}^m + \text{FN}^m + \text{FP}^m) \times 100\%$, $\text{sns/rcl}(\%) = (\text{TP}^m/\text{TP}^m + \text{FN}^m) \times 100\%$, $\text{spc}(\%) = (\text{TN}^m/\text{TN}^m + \text{FP}^m) \times 100\%$, $\text{prc}(\%) = (\text{TP}^m/\text{TP}^m + \text{FP}^m) \times 100\%$, and $F1 - \text{score}(\%) = (2 \times \text{prc} \times \text{rcl}/\text{prc} + \text{rcl}) \times 100\%$, where TP^m, TN^m, FP^m, and FN^m denote the true positive, true negative, false positive, and false negative of mth class, respectively and can be obtained as [8]: $\text{TP}^m = C_{m,m}$, $\text{FN}^m = \sum_{n=1,n\neq m}^{7} C_{n,m}$, $\text{FP}^m = \sum_{n=1,n\neq m}^{7} C_{m,n}$, $\text{TN}^m = \sum_{n=1}^{7}\sum_{k=1}^{7} C_{n,k} - \text{TP}^m - \text{FP}^m - \text{FN}^m$.

B. Performance Evaluation

The performance of the proposed Pulmo-TS2ONN is assessed by training and testing the model using a fivefold cross-validation approach. Initially, all RSs from the three different databases were combined and then split into an 80%–20% training and testing set. Additionally, we generate the validation set using 10% of the data from the training set. The optimal simulation parameters used to train the proposed Pulmo-TS2ONN are shown in Table IV, which have been selected using the GridSearchCV-based hyperparameter optimization framework [41]. Fig. 6(a) and (b) shows the

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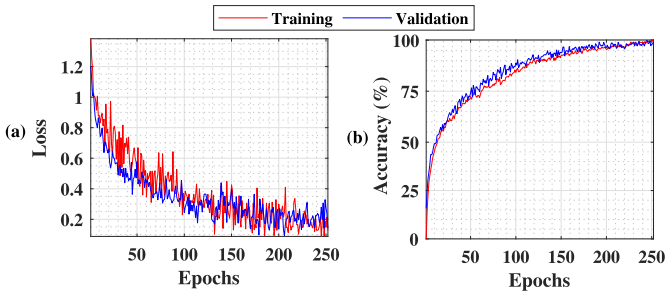


Fig. 6. (a) and (b) Loss and accuracy curve of proposed Pulmo-TS2ONN for seven class PD detection.

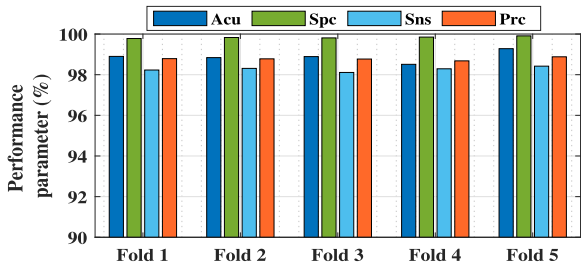


Fig. 7. Fold-wise variation of performance parameters.

TABLE V

CLASS-SPECIFIC PERFORMANCE PARAMETERS OBTAINED THROUGH FIVEFOLD CROSS-VALIDATION OF THE PULMO-TS2ONN MODEL

PD classes	Performance parameters (mean ± standard deviation)			
	Rcl (%)	Spc (%)	Prc (%)	F1-score (%)
Asthma	98.69 ± 0.90	99.59 ± 0.19	98.69 ± 1.01	98.67 ± 0.55
BRON	98.23 ± 0.87	99.93 ± 0.06	98.84 ± 1.05	98.54 ± 1.08
Bronchiolitis	97.39 ± 1.01	99.90 ± 0.05	98.19 ± 0.94	97.79 ± 1.01
COPD	99.37 ± 0.34	99.84 ± 0.14	99.68 ± 0.28	99.52 ± 0.29
Healthy	99.86 ± 0.58	99.83 ± 0.12	99.26 ± 0.40	99.06 ± 0.39
Pneumonia	99.67 ± 0.44	99.80 ± 0.15	97.99 ± 0.94	98.82 ± 0.51
URTI	95.76 ± 0.81	99.90 ± 0.11	98.84 ± 1.72	97.27 ± 1.97

loss and accuracy curves of our proposed Pulmo-TS2ONN model for both the training and validation phases. It can be observed from Fig. 6(b), that the Pulmo-TS2ONN model achieves excellent accuracy without overfitting the model.

To illustrate the supremacy of the proposed Pulmo-TS2ONN framework, we have also provided the fold-wise classification results in Fig. 7. It can be seen from the figure, that our model archives high accuracy rates in each of the folds. The average acu, spc, sns, and prc achieved by the Pulmo-TS2ONN are 98.88%, 99.84%, 98.27%, and 98.78%, respectively.

To assess the effectiveness of the class-specific classification, we also performed a statistical analysis of the class-wise performance parameter calculation. Table V illustrates the variation in classification metrics for each of the individual classes of PDs obtained from the fivefold cross-validation of the proposed Pulmo-TS2ONN model.

We have also provided the class-wise receiver operator characteristics (ROC) curve obtained by testing the Pulmo-TS2ONN model using the test TTFFS, in Fig. 8. It can be observed that each of the individual ROC curves yields a high area under the curve (AUC) value. This, in a way, represents that the proposed Pulmo-TS2ONN model provides high classification performance for each of the disease classes.

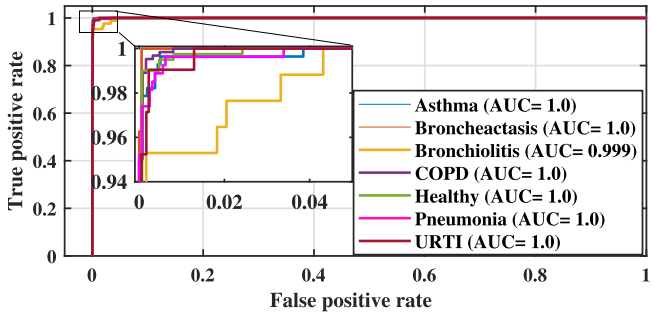


Fig. 8. ROC curve obtained from the proposed Pulmo-TS2ONN model.

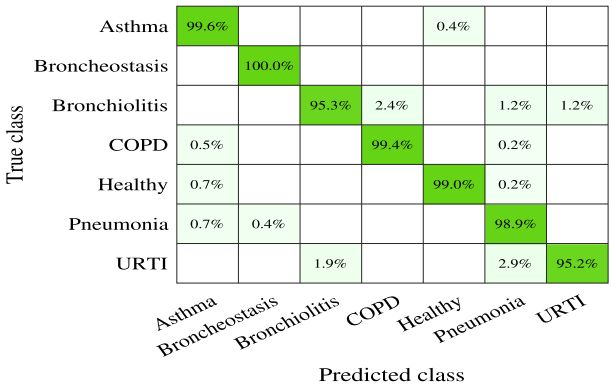


Fig. 9. Illustrates normalized confusion matrix for seven-class PD classification using Pulmo-TS2ONN.

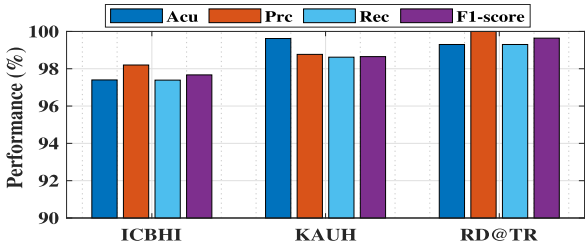


Fig. 10. Database-wise classification performance using the proposed Pulmo-TS2ONN framework.

To prove the efficacy of our proposed framework, we have provided the normalized confusion matrix of the seven-class PD classification in Fig. 9. From Fig. 9, it is evident that the misclassification rate for each of the classes is very low.

Further, we have provided the database-wise classification performance in Fig. 10 which illustrates that the proposed framework archives decent performance for each of the databases. From Fig. 10, it can be seen that the Pulmo-TS2ONN achieves an accuracy of 97.40%, 99.62%, and 99.62% for ICBHI, KAUH, and RD@TR databases.

Additionally, Fig. 11(a) and (b) illustrates the performance of the proposed framework in terms of efficient class-specific feature extraction using a 2-D t-distributed stochastic neighbor embedding (t-SNE) [42] visualization. We have projected both the raw TTFFS and the SONS-driven concatenated embeddings (E_{concat}) to the 2-D-t-SNE plane in Fig. 11(a) and (b), respectively. It can be seen that initially, all the raw TTFFS features from different disease classes were randomly scattered in the 2-D t-SNE plane; however, our proposed Pulmo-TS2ONN

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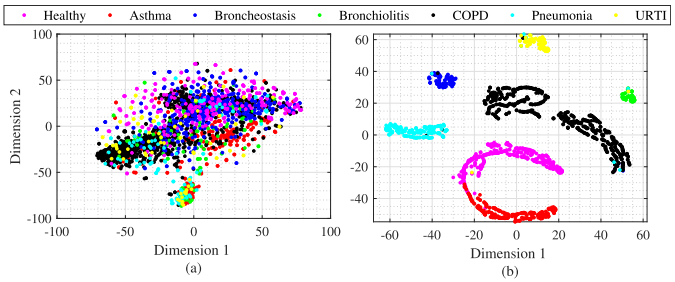


Fig. 11. 2-D-tSNE feature plot of (a) Raw TTFFS and (b) SONS-driven concatenated embeddings (E_{concat}).

TABLE VI
QUANTITATIVE PERFORMANCE COMPARISON OF THE PROPOSED PULMO-TS2ONN WITH OTHER BAS

Model architecture	TFRs utilized	Required SONS	Evaluation metrics (%)			
			Acu	Rcl	prc	F1 score
BA 1	Mel spectrogram	1	88.13	82.37	90.02	86.03
BA 2	MFCC		78.99	78.66	79.86	79.26
BA 3	CQT		89.97	89.41	86.52	87.94
BA 4	CQT, MFCC	2	54.41	36.88	56.01	44.48
BA 5	Mel spectrogram, CQT		60.86	64.27	42.59	51.23
BA 6	Melspectrogram, MFCC		83.5	79.43	85.21	82.22
Pulmo-TS2ONN	Melspectrogram, MFCC, CQT	3	**98.88**	**98.27**	**98.78**	**98.52**

is able to extract distinct feature representation from each class, and as a result, it aids in the formation of distinct clusters in the 2-D t-SNE plane.

C. Ablation Study of Different Impacting Design Parameters

This section includes the impact of various design parameter choices that affect the overall classification performance of the Pulmo-TS2ONN framework.

1) Selection of Optimal Number of SONS Blocks and TFRs: The SONS blocks that extract compact feature representations from the TTFFS, are one of the key backbone DL blocks of our proposed Pulmo-TS2ONN architecture. In this article, we have utilized three TFRs in conjunction with three SONS blocks to construct the TS2ONN architecture. However, finding an optimal number of SONS blocks and the optimal choice of TFR becomes an essential factor. To study the aforementioned aspects, we created six baseline architecture (BA) designs, three (BA 1, 2, 3) of which employ a single TFR and a single SONS, while the other three (BA 4, 5, 6) use two TFRs and two SONSs. To ensure a fair comparison, we kept the architectural specifications of the SONS same across all BAs and used a fivefold cross-validation-based training-testing strategy to compare their performance with our proposed Pulmo-TS2ONN in Table VI. Table VI demonstrates that using TTFFs with three SONS block-based Pumo-TS2ONN architecture yields the highest classification rates compared to other BA architectures that use either a single TFR or a combination of two TFRs. Additionally, to illustrate the efficacy of the Pulmo-TS2ONN, a 2-D-tSNE feature visualization for each of the BAs has been shown in Fig. 12(a)–(f), respectively. As we compare the t-SNE feature visualization performance of Pulmo-TS2ONN shown in Fig. 11(b) with the other t-SNEs

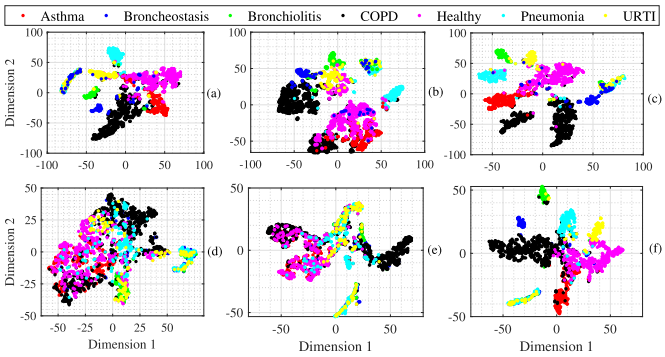


Fig. 12. 2-D t-SNE visualization of features extracted from (a) BA 1, (b) BA 2, (c) BA 3, (d) BA 4, (e) BA 5, and (f) BA 6, respectively.

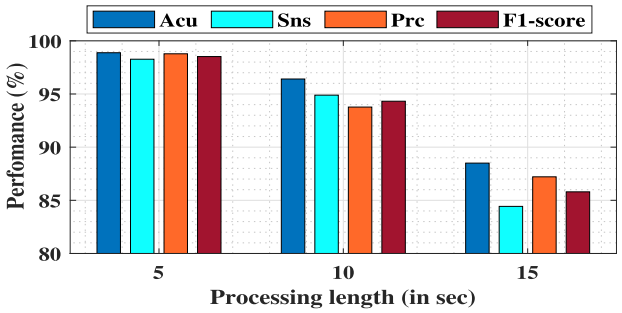


Fig. 13. Effect of processing length of input RS on overall classification performance of the Pulmo-TSONN framework.

of the BAs shown in Fig. 12(a)–(f), it is evident that Pulmo-TS2ONN outperforms the BAs in terms of the creation of class-specific clusters.

2) Selection of Optimal RS Processing Length: Due to the presence of nonstationary spectral components in RSs, many researchers focus on employing TFRs for RS classification. However, as the RS features appear in the RS time series sequentially, it is crucial to ascertain how the processing length affects classification performance. To illustrate the effect of processing length on the classification performance we have provided a comparative study in Fig. 13(a). We have uniformly varied the processing length from 5 to 15 s and noted the classification metrics in each case. From Fig. 13, it can be seen that with 5 s processing length, the proposed Pulmo-TS2ONN yields the highest accuracy, however, with the increase in the processing length, the performance of the Pulmo-TS2ONN deteriorates drastically.

D. Performance Evaluation on Real-Time Data

In this section, we examine the performance of the proposed framework on real-time RS data collected from the subjects affected with PD, admitted to the pulmonary medicine department of All India Institute of Medical Sciences Patna. Due to the limited availability of patients, we have acquired RSs of 20 s length from two COPD (annotated with PID-01, and 02), and two asthmatic patients (PID-03, and 04) using Littmann 3200 digital stethoscope and transferred the signals to a laptop over Bluetooth using the Littmann[1] StethAssist[2]

[1]Registered trademark.
[2]Trademarked.

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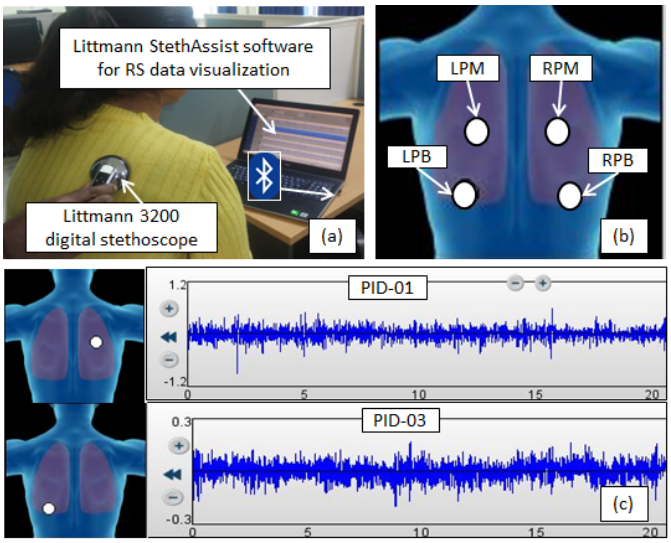


Fig. 14. Illustrates (a) experimental set-up of the RS data acquisition, (b) acquisition sites, and (c) visualization of RS data collected from PID-01, 03.

TABLE VII
CLASSIFICATION PERFORMANCE ON REAL-TIME ACQUIRED RS DATA

Classes (no. of recordings)	Acquisition sites	acu (%)	Acquisition site (avg. acu in %)
Healthy (24)	Left posterior middle (LPM), right posterior middle (RPM), left posterior base (LPB), right posterior base (RPB)	Overall (98.56) Healthy (97.61) Asthma (100) COPD (100)	LPM(100), RPM(97.62), LPB(97.62), RPB(95.24)
Asthma (3)			RPM(100), LPB (100), RPB (100)
COPD (5)			RPM (100), RPB (100), LPM (100)

software. Additionally, we have collected RS signals from six healthy subjects (PID-05 to PID-10). Fig. 14(a) illustrates the experimental setup, Fig. 14(b) depicts the different RS acquisition sites, and Fig. 14(c) shows the visualization of RS data collected from PID-01 and PID-03, respectively. The classification performance of the proposed Pulmo-TS2ONN on the real-time recorded RS data is tabulated in Table VII.

E. On-Board Implementation of Pulmo-TS2ONN

The implementation of this PD classification algorithm on a low-resource embedded processor not only opens up possibilities for Internet of Medical Things (IoMT) applications [43], but it also enables the direction of developing personalized respiratory health monitoring systems [44]. However, the large parameter size and high computational complexity hinder the deployability of these DL-based frameworks on a low-resource microcontroller [44]. In this article, the proposed Pulmo-TS2ONN framework has been successfully implemented on an ARM-A7 quad-core Broadcom BCM2711 microcontroller, popularly known as the Raspberry Pi-4, which runs at 1.5 GHz clock and comprises 4 GB of RAM. Fig. 15(a) illustrates the experimental setup of the microcontroller, and Fig. 15(b)–(d) shows the sample classification results of Pulmo-TS2ONN. The end-to-end PD classification pipeline was first trained on the Google Colab Pro+ platform and then subsequently, the model weights were transferred to the microcontroller. In addition to that, necessary libraries such as numpy, librosa,

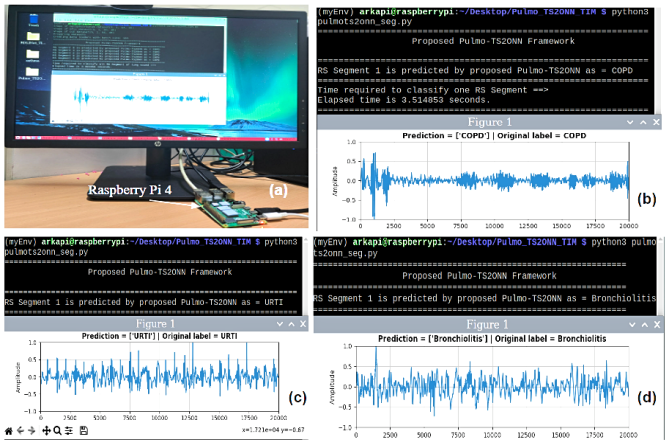


Fig. 15. (a) On-board implementation setup of Pulmo-TS2ONN using Raspberry Pi-4 microcontroller and (b)–(d) illustrates the sample classification results.

TABLE VIII
COMPARATIVE RESULTS OF THE PROPOSED PULMO-TS2ONN MODEL WITH FINE-TUNED DEEP TRANSFER LEARNING MODELS

Model	Model judgementing parameter					
	Parameters (million (M))	FLOPs (Gillion (G))	Performance measures (%)			
			acu	sns	prc	F1-score
ResNet-50	25.624	4.1437	90.86	90.48	92.66	91.55
MobileNet	4.232	0.5272	92.33	93.43	93.55	93.49
Shuffle Net	5.468	0.4905	91.4	89.3	91.36	90.63
Pulmo-TS2ONN	**0.414**	**0.0055**	**98.88**	**98.27**	**98.78**	**98.52**

torch, etc., were also installed in the microcontroller to run the developed Pulmo-TS2ONN algorithm. Upon experimentation, it has been seen that the Pulmo-TS2ONN requires a latency of 3.48 ± 0.27 s and a peak memory requirement of 15.2924 ± 0.0018 MB to categorize one RS segment. This indicates that the proposed framework can be translated into a real-life clinical scenario for PD screening using the RS measurements.

F. Performance Comparison

1) Comparison With SOTA Fine-Tuned Transfer Learning DL Frameworks: We have compared our proposed framework with the existing transfer learning-based DL models such as ResNet-50, MobileNet, and ShuffleNetV2, which have been re-trained using the fine-tuning mechanism to classify seven PD categories. Table VIII depicts the comparative study of the proposed Pulmo-TS2ONN model with different fine-tuned transfer learning models in terms of several model assessment metrics. From Table VIII, it is evident that our SONS-driven Pulmo-TS2ONN model outperforms each of the fine-tuned transfer learning models by achieving the highest classification rates. Therefore, it can also be argued that the use of enhanced nonlinear GOPM helps to attain such higher classification performance than the existing CNN models, where the primary neural backbone is the linear perceptron model. Additionally, Table VIII also illustrates that the parameter size and the total number of floating point operations required (FLOPs) for the proposed Pulmo-TS2ONN model are 0.414 million (M) and 0.0055 gillion (G) only, which are significantly less than any of the fine-tuned transfer learning models.

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TABLE IX
PERFORMANCE COMPARISON OF PROPOSED PULMO-TS2ONN WITH EXISTING SOTA TECHNIQUES ON RS-BASED PD CLASSIFICATION

Sl. No.	Reference	Preprocessing	Features	Classification technique	Number of classes	Database	Performance parameters (%)		
							Acu (%)	Sns (%)	Spc (%)
1	Basu et al. [19]	Framing into 10 sec window	MFCC	GRU	6	ICBHI	95.7	-	-
2	Subho et al. [16]	Framed into 5sec segment, EMD based best IMF selection	Scalogram	LCNN	6	ICBHI	98.7	98.96	100
3	Tripathy et al. [21]	Amplitude normalization, EWT decomposition	Spectro (peak amplitude, peak frequency)-temporal features (Shannon entropy)	MLP, SVM, LGBM	4	KAUH	80.35	84.88	75.23
4	Shi et al. [17]		Mel spectrogram	Multi-Resolution Interleaved Net	4	ICBHI+ Respiratory Database@TR	91.56	–	88.73
5	Roy et al. [8]	Resampling at 4 kHz, BWC elimination, amplitude normalization	Mel spectrogram	RDLINet	7	ICBHI+ KAUH + Respiratory Database@TR	96.6	96.2	98
6	**Proposed work**	**Resampling at 4 kHz, BWC elimination, and amplitude normalization**	**Mel spectrogram, CQT, MFCC based TTFFS**	**Pulmo-TS2ONN**	**7**	**ICBHI+ KAUH + Respiratory Database@TR**	**98.88**	**98.27**	**99.84**

2) Comparison With SOTA Techniques on RS-Based PD Classification: In this section, we compare our proposed Pulmo-TS2ONN framework with some noteworthy prior works on RS-based PD classification. Table IX illustrates the comparative performance of the Pulmo-TS2ONN with other existing works on RS-based disease detection using different publicly available databases [9], [10], [12]. Most of the works have considered a single database for multiclass PD classification (either four-class or six-class). In the case of ICBHI database, Basu and Rana [19], and Shuvo et al. [16] achieved decent accuracy on six-class disease classification; however, Shi et al. [17] achieved 91.56% accuracy when the RS data of ICBHI and RespiratoryDatabase@TR was combined for four-class classification. While performing the same four-class classification with KAUH database, Tripathy et al. [21] achieved an inferior accuracy rate. Roy and Satija et al. [8] only performed seven-class classification by employing all three databases. However, it is evident from Table IX that the proposed Pulmo-TS2ONN framework surpasses all the existing SOTA techniques on RS-based seven-class PD detection by achieving the highest accuracy of 98.88%.

G. Performance Comparison of on-Device Implementation

To illustrate the efficacy of the proposed framework in terms of deployability on low-compute hardware, we have implemented different DL-based PD classification pipelines, such as LCNN [16], RDLINet [8], etc., for seven-class PD classification task and implemented on Raspberry pi-4. We have measured the peak memory usage and latency to classify one RS segment and presented the same in Table X. We can see that both latency and peak memory usage for LCNN [16] are high, as the iterative-EMD-based scalogram generation process is time-consuming. Additionally, we can observe that our Pulmo-TS2ONN requires slightly higher memory and latency than our previously published RDLINet [8]. This behavior is obvious as Pulmo-TS2ONN requires a total of three TFR extraction and the model parameter size is higher than RDLINet [8]. In contrast to RDLINet, the Pulmo-TS2ONN demonstrates a notable enhancement in overall accuracy (with a 2.28%) as well as class-wise PD classification accuracy, which specifically adds a substantial clinical value to enhance diagnosis and enable accurate medication recommendations [22]. Thus, despite the trade-offs of somewhat longer processing time and peak memory utilization compared to RDLINet, the cumulative therapeutic advantages considerably surpass these concerns.

TABLE X
PERFORMANCE COMPARISON OF THE PROPOSED PULMO-TS2ONN FRAMEWORK WITH EXISTING WORKS BASED ON THE PARAMETERS OBTAINED AFTER ON-DEVICE IMPLEMENTATION

DL model	Model size	Acu (%)	Latency/ inference time (s)	Peak memory usage (MB)
LCNN [16]	44.85 MB	93.88	50.53 ± 0.41	207.1331 ± 0.0079
RDLINet [8]	498 KB	96.6	2.97 ± 0.18	6.5393 ± 0.0038
Pulmo-TS2ONN	2.02 MB	98.88	3.48 ± 0.27	15.2924 ± 0.0018

V. CONCLUSION

In this article, we investigate the potential of the TS2ONN architecture in conjunction with the fusion of the triple TFR-based TTFFS extracted from RSs. By utilizing our proposed Pulmo-TS2ONN framework, we surpassed the SOTA algorithms for RS-based seven-class PD detection utilizing three publicly accessible databases, achieving the highest accuracy of 98.88%. The suggested framework also demonstrates the advantages of TS2ONN architecture over vanilla pretrained CNNs that employ basic linear perceptron design by attaining higher classification rates with fewer parameters. Our future research will focus on the impact of different factors, such as the presence of heart sound interference and ambient noise in RS, that may have a detrimental effect on PD classification and how to suppress these noise elements to obtain better outcomes. Furthermore, the successful deployment of the proposed Pulmo-TS2ONN on a Raspberry Pi-4 microcontroller shows the possibility of developing an on-device point-of-care system for diagnosing and screening various PDs in real-world clinical situations based on RSs.

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REFERENCES

[1] S. M. Levine and D. D. Marciniuk, "Global impact of respiratory disease: What can we do, together, to make a difference?" *Chest*, vol. 161, no. 5, pp. 1153–1154, 2022.

[2] J. L. Schneider, J. H. Rowe, C. Garcia-de-Alba, C. F. Kim, A. H. Sharpe, and M. C. Haigis, "The aging lung: Physiology, disease, and immunity," *Cell*, vol. 184, no. 8, pp. 1990–2019, Apr. 2021.

[3] A. A. Cruz, *Global Surveillance, Prevention and Control of Chronic Respiratory Diseases: A Comprehensive Approach*. Geneva, Switzerland: World Health Organization, 2007.

[4] S. Trivedy, M. Goyal, P. R. Mohapatra, and A. Mukherjee, "Design and development of smartphone-enabled spirometer with a disease classification system using convolutional neural network," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 9, pp. 7125–7135, Sep. 2020.

[5] A. B. Bhome, "COPD in India: Iceberg or volcano?" *J. Thoracic Disease*, vol. 4, no. 3, p. 298, 2012.

[6] B. Roy, A. Roy, J. K. Chandra, and R. Gupta, "I-PRExT: Photoplethysmography derived respiration signal extraction and respiratory rate tracking using neural networks," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–9, 2021.

[7] National Institutes of Health, National Asthma Education and Prevention Program, *Expert Panel Report III: Guidelines for the Diagnosis and Management of Asthma*, Bethesda, MD, USA, 1997.

[8] A. Roy and U. Satija, "RDLINet: A novel lightweight inception network for respiratory disease classification using lung sounds," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–13, 2023.

[9] B. Rocha et al., "A respiratory sound database for the development of automated classification," in *Proc. Int. Conf. Biomed. Health Informat.* Singapore: Springer, 2017, pp. 33–37.

[10] M. Fraiwan, L. Fraiwan, B. Khassawneh, and A. Ibnian, "A dataset of lung sounds recorded from the chest wall using an electronic stethoscope," *Data Brief*, vol. 35, Apr. 2021, Art. no. 106913.

[11] L. Pham, H. Phan, R. Palaniappan, A. Mertins, and I. McLoughlin, "CNN-MoE based framework for classification of respiratory anomalies and lung disease detection," *IEEE J. Biomed. Health Informat.*, vol. 25, no. 8, pp. 2938–2947, Aug. 2021.

[12] G. Altan, Y. Kutlu, Y. Garbi, A. Ö. Pekmezci, and S. Nural, "Multimedia respiratory database (RespiratoryDatabase@TR): Auscultation sounds and chest X-rays," *Natural Eng. Sci.*, vol. 2, no. 3, pp. 59–72, Oct. 2017.

[13] L. Silva et al., "COVID-19 respiratory sound analysis and classification using audio textures," *Frontiers Signal Process.*, vol. 2, Oct. 2022, Art. no. 986293.

[14] J. Acharya and A. Basu, "Deep neural network for respiratory sound classification in wearable devices enabled by patient specific model tuning," *IEEE Trans. Biomed. Circuits Syst.*, vol. 14, no. 3, pp. 535–544, Jun. 2020.

[15] T. Nguyen and F. Pernkopf, "Lung sound classification using co-tuning and stochastic normalization," *IEEE Trans. Biomed. Eng.*, vol. 69, no. 9, pp. 2872–2882, Sep. 2022.

[16] S. B. Shuvo, S. N. Ali, S. I. Swapnil, T. Hasan, and M. I. H. Bhuiyan, "A lightweight CNN model for detecting respiratory diseases from lung auscultation sounds using EMD-CWT-based hybrid scalogram," *IEEE J. Biomed. Health Informat.*, vol. 25, no. 7, pp. 2595–2603, Jul. 2021.

[17] L. Shi, J. Zhang, B. Yang, and Y. Gao, "Lung sound recognition method based on multi-resolution interleaved net and time-frequency feature enhancement," *IEEE J. Biomed. Health Informat.*, vol. 27, no. 10, pp. 4768–4779, Oct. 2023.

[18] D. Perna, "Convolutional neural networks learning from respiratory data," in *Proc. IEEE Int. Conf. Bioinf. Biomed. (BIBM)*, Dec. 2018, pp. 2109–2113.

[19] V. Basu and S. Rana, "Respiratory diseases recognition through respiratory sound with the help of deep neural network," in *Proc. 4th Int. Conf. Comput. Intell. Netw. (CINE)*, Feb. 2020, pp. 1–6.

[20] M. T. García-Ordás, J. A. Benítez-Andrades, I. García-Rodríguez, C. Benavides, and H. Alaiz-Moretón, "Detecting respiratory pathologies using convolutional neural networks and variational autoencoders for unbalancing data," *Sensors*, vol. 20, no. 4, p. 1214, Feb. 2020.

[21] R. K. Tripathy, S. Dash, A. Rath, G. Panda, and R. B. Pachori, "Automated detection of pulmonary diseases from lung sound signals using fixed-boundary-based empirical wavelet transform," *IEEE Sensors Lett.*, vol. 6, no. 5, pp. 1–4, May 2022.

[22] A. Mondal, P. Bhattacharya, and G. Saha, "Detection of lungs status using morphological complexities of respiratory sounds," *Sci. World J.*, vol. 2014, pp. 1–9, Mar. 2014.

[23] A. Roy and U. Satija, "AsTFSONN: A unified framework based on time-frequency domain self-operational neural network for asthmatic lung sound classification," in *Proc. IEEE Int. Symp. Med. Meas. Appl. (MeMeA)*, Jun. 2023, pp. 1–6.

[24] M. U. Zahid, S. Kiranyaz, and M. Gabbouj, "Global ECG classification by self-operational neural networks with feature injection," *IEEE Trans. Biomed. Eng.*, vol. 70, no. 1, pp. 205–215, Jan. 2023.

[25] M. Sarkar, I. Madabhavi, N. Niranjan, and M. Dogra, "Auscultation of the respiratory system," *Ann. Thoracic Med.*, vol. 10, no. 3, p. 158, 2015.

[26] S. A. H. Tabatabaei, P. Fischer, H. Schneider, U. Koehler, V. Gross, and K. Sohrabi, "Methods for adventitious respiratory sound analyzing applications based on smartphones: A survey," *IEEE Rev. Biomed. Eng.*, vol. 14, pp. 98–115, 2021.

[27] S. Reichert, R. Gass, C. Brandt, and E. Andrès, "Analysis of respiratory sounds: State of the art," *Clin. Med. Circulatory, Respiratory Pulmonary Med.*, vol. 2, Jan. 2008, Art. no. CCRPM.S530.

[28] S. A. F. Jaffery, S. Aziz, M. U. Khan, S. Z. H. Naqvi, M. Faraz, and A. Usman, "An automated system for the classification of bronchiolitis and bronchiectasis diseases using lung sound analysis," in *Proc. Int. Conf. Robot. Autom. Ind. (ICRAI)*, Mar. 2023, pp. 1–6.

[29] U. Satija, B. Ramkumar, and M. S. Manikandan, "Real-time signal quality-aware ECG telemetry system for IoT-based health care monitoring," *IEEE Internet Things J.*, vol. 4, no. 3, pp. 815–823, Jun. 2017.

[30] B. Bahmei, E. Birmingham, and S. Arzanpour, "CNN-RNN and data augmentation using deep convolutional generative adversarial network for environmental sound classification," *IEEE Signal Process. Lett.*, vol. 29, pp. 682–686, 2022.

[31] I. Ozer, "Pseudo-colored rate map representation for speech emotion recognition," *Biomed. Signal Process. Control*, vol. 66, Apr. 2021, Art. no. 102502.

[32] J. C. Brown, "Calculation of a constant Q spectral transform," *J. Acoust. Soc. Amer.*, vol. 89, no. 1, pp. 425–434, Jan. 1991.

[33] B. Gfeller, C. Frank, D. Roblek, M. Sharifi, M. Tagliasacchi, and M. Velimirovic, "SPICE: Self-supervised pitch estimation," *IEEE/ACM Trans. Audio, Speech, Language Process.*, vol. 28, pp. 1118–1128, 2020.

[34] Z. K. Abdul and A. K. Al-Talabani, "Mel frequency cepstral coefficient and its applications: A review," *IEEE Access*, vol. 10, pp. 122136–122158, 2022.

[35] S. Kiranyaz, T. Ince, A. Iosifidis, and M. Gabbouj, "Operational neural networks," *Neural Comput. Appl.*, vol. 32, no. 11, pp. 6645–6668, Jun. 2020.

[36] S. Kiranyaz, J. Malik, H. B. Abdallah, T. Ince, A. Iosifidis, and M. Gabbouj, "Self-organized operational neural networks with generative neurons," *Neural Netw.*, vol. 140, pp. 294–308, Aug. 2021.

[37] M. Lin, Q. Chen, and S. Yan, "Network in network," 2013, *arXiv:1312.4400*.

[38] D. Hendrycks and K. Gimpel, "Gaussian error linear units (GELUs)," 2016, *arXiv:1606.08415*.

[39] S. Das, D. Jyotishi, and S. Dandapat, "Automated detection of heart valve diseases using stationary wavelet transform and attention-based hierarchical LSTM network," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–10, 2023.

[40] T. Kautz, B. M. Eskofier, and C. F. Pasluosta, "Generic performance measure for multiclass-classifiers," *Pattern Recognit.*, vol. 68, pp. 111–125, Aug. 2017.

[41] M. Saini, U. Satija, and M. D. Upadhayay, "DSCNN-CAU: Deep-learning-based mental activity classification for IoT implementation toward portable BCI," *IEEE Internet Things J.*, vol. 10, no. 10, pp. 8944–8957, May 2023.

[42] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 1–27, 2008.

[43] M. Saini, D. Kumar, and U. Satija, "Edge of medical things implementation for deep learning-based cognitive task recognition," *IEEE Internet Things Mag.*, vol. 5, no. 3, pp. 56–60, Sep. 2022.

[44] D. Pal, S. Mukhopadhyay, and R. Gupta, "Two-stage classifier for resource constrained on-board cardiac arrhythmia detection," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–10, 2023.

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